

Equivalent definitions: linear independence

1) no x_i expressible as a superposition of $\{x_1, \dots, x_n\} \setminus \{x_i\}$

2) there are not a_i , not all zero, s.t.

$$0 = a_1 x_1 + a_2 x_2 + \dots + a_n x_n$$

3) the only solution of $0 = a_1 x_1 + \dots + a_n x_n$ is

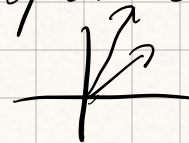
$$a_i = 0 \quad \forall i$$

Lem: any set containing the zero vector is linearly dependent

any list containing the same vector twice is linearly dependent

Ex: is $\{(1,2), (3,4)\} \subset \mathbb{R}^2$ linearly independent?

yes, $\begin{matrix} 1 = a_3 \\ 2 = a_4 \end{matrix} \times$ not possible



or

$$\begin{aligned} 0 &= (a+3b, 2a+4b) \\ 0 &= a+3b \\ 0 &= 2a+4b \end{aligned}$$

Ex: is $\{(1,i), (6i,-6)\} \subset \mathbb{C}^2$ linearly independent?

no, $(6i, -6) = 6i(1, i)$

Q: What is the minimal amount of info abt a linear function so that we know its value on all inputs

Suppose $L: A \rightarrow B$ is linear

• if L is known on $\{x_1, \dots, x_n\}$ and $\text{span}\{x_1, \dots, x_n\} = A$ this is sufficient

$$w \in A \Rightarrow w \in \text{span}\{x_1, \dots, x_n\} \Rightarrow w = a_1 x_1 + \dots + a_n x_n$$

$$L(w) = L(a_1 x_1 + \dots + a_n x_n) = a_1 L(x_1) + \dots + a_n L(x_n)$$

- if $\{x_1, \dots, x_n\}$ is also linearly independent then it is the minimal amount of info.
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Lem: if $\{x_1, \dots, x_n\}$ is linearly dependent then there is x_i s.t. $\text{span}\{x_1, \dots, x_n\} \setminus \{x_i\} = \text{span}\{x_1, \dots, x_n\}$

Ex: $\{(1, 2), (3, 4), (1, -6)\}$ is linearly dependent
that is, $\text{span}\{(1, 2), (3, 4)\} = \text{span}\{(1, 2), (3, 4), (1, -6)\}$